

1320484 - Stored Procedures in SAP Business One

Component: SBO-SDK-TNSP (SAP Business One > Software Development Kit > SDK Transaction Notification Stored procedure), Version: 28, Released On: 25.04.2025

Symptom

This SAP Note deals with SAP Business One stored procedures, their purposes, usage, limitations and support scope.

Solution

Purpose:

The purpose of *Transaction Notifications* is to enable the partner/customer to receive notification and register logs on any transaction taking place within the SAP Business One database. SAP Business One provides two stored procedure mechanisms for receiving notification of data-driven events.

- **SBO_SP_TransactionNotification**
- **SBO_SP_PostTransactionNotice**

It is triggered for the majority of the SAP Business One transactions and user-defined object (UDO), whenever operations such as add, update, delete, cancel, close are performed in SAP Business One Client or Data Interface (DI) API or DI Server or Service Layer.

SBO_SP_TransactionNotification is triggered **before** the SAP Business One transaction is committed. It can be used to roll back transactions and display custom error messages in the SAP Business One Client.

SBO_SP_PostTransactionNotice is triggered **after** the SAP Business One transaction is committed.

Depending on the Business Object and the operation performed, there could be multiple notification procedures triggered in a sequence. For example, when you update an item code, both notifications will be triggered for every warehouse associated to the item code and also for the item code itself.

- **SBO_SP_EDF_TransactionNotification(Obsolete)**

This was introduced for Electronic document handling, but no longer has usage.

Usage:

Notification Procedures can be edited by the user to add SQL scripts as part of a notification process.

Both stored procedures have 5 parameters.

Parameter	Example	Remarks
@objecttype	2 (Business Partner)	A full list of object types exposed through DI API can be found in the DI API documentation under the section <i>Enumerations</i> . The enumeration is called BoObjectTypes .
@transaction_type	U (update)	It may have values of: • A (where a record was added) • U (where a record was updated) • D (where a record was deleted) • C (where a document was canceled) • L (where a document was closed)
@num_of_cols_in_key	1	Returns the number of columns or fields in the key to the record. • A Business Partner, for example, has a key consisting of a single field (<i>CardCode</i>), so this variable would have a value of "1". • A Special Prices object, however, has a key consisting of the <i>CardCode</i> and <i>ItemCode</i>, so for a Special Prices object, this variable would have a value of "2".
@list_of_key_cols_tab_del	CardCode	It returns a tab-delimited list of column names (field names) that represent the object key. • For a Business Partner, this would be <i>CardCode</i>. • For a Special Prices object, it would be <i>CardCode ItemCode</i>.

@list_of_cols_val_tab_del	C40001	This returns a tab-delimited list of values required to retrieve the object from SAP Business One. - For example, a Business Partner record might have a value of "C40001". - A Special Prices object would contain 2 fields separated by a tab character, for example: "V10005 ... A00001"
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Both stored procedures have 2 return values.

@error - You can set any value other than 0 to trigger rollback of a transaction (SBO_SP_TransactionNotification) or register the transaction on the log files (SBO_SP_PostTransactionNotice).

@error_message - You set a string value which will be displayed in the SAP Business One Client status bar (SBO_SP_TransactionNotification) or on the log files (SBO_SP_PostTransactionNotice).

You can add your validation as below.

Sample code for MSSQL

```

Alter proc SBO_SP_TransactionNotification
@object_type nvarchar(20),-- SBO Object Type
@transaction_type nchar(1),-- [A]dd, [U]pdate, [D]elete, [C]ancel, [L]ose
@num_of_cols_in_key int,
@list_of_key_cols_tab_del nvarchar(255),
@list_of_cols_val_tab_del nvarchar(255)
AS
begin
-- Return values
declare @error int -- Result (0 for no error)
declare @error_message nvarchar (200)-- Error string to be displayed
select @error = 0
select @error_message = N'Ok'

-----

-- ADD YOUR CODE HERE

-----

IF(@object_type='224')
begin
if(@transaction_type = 'U' or @transaction_type = 'A' or @transaction_type = 'D')
begin
--Set error value other than 0 to rollback transactions
select @error = 1001
--Set error message,which will be shown in the Business one client
select @error_message = N'You are not allowed to change the translation'
end
end
-- Select the return values
select @error, @error_message
end

```

The additional code must be exception free. No TRY/CATCH handling in MSSQL Database will be supported.

Enhancements:

- **Enable/Disable TransactionNotification:**

By default, *TransactionNotification* is enabled and triggered for the majority of SAP Business One transactions.

Starting from *SAP Business One 9.1 PLO5*, TransactionNotification can be disabled for a company database for **all** objects (*Administration* → *System Initialization* → *General Setting* → *Service*).

TransactionNotification can also be disabled for a **specific** object by setting an entry in the [CompanyBD].CTNS table. Currently there is no user interface in SAP Business One to set the entries. They need to be set manually as below.

INSERT INTO CTNS VALUES ((select max(Absid) + 1 from CTNS),'2','Y','N','N');

Field	Description	Type	Size	Related	Default Value	Constraints	
AbsId	Internal No.	Int	11	-			
ObjectId	Business Object ID	nVarChar	20	-			
EnableAsyN	Enable async notification	VarChar	1	-	N	Y	Yes
						N	No
EnableTn	Transaction Notification	VarChar	1	-	Y	Y	Yes
						N	No
EnablePTn	Post Transaction Notification	VarChar	1	-	Y	Y	Yes
						N	No

CTNS.EnableAsyN = 'Y' means that relevant operations on this object will trigger SEVT notification; 'N' means they will not

CTNS.EnableTn = 'Y' means that relevant operations on this object will trigger an SBO_SP_TRANSACTIONNOTIFICATION notification; 'N' means they will not

CTNS.EnablePTn = 'Y' means that relevant operations on this object will trigger an SBO_SP_POSTTRANSACTIONNOTICE notification; 'N' means they will not

By default, the following objects are registered into CTNS to **NOT** send the three kinds of notifications (SEVT; SBO_SP_TRANSACTIONNOTIFICATION; SBO_SP_POSTTRANSACTIONNOTICE):

- 10000044 (BTN)
- 10000045 (SRN)
- 10000066 (SRQ)
- 10000067 (BTQ)
- 310000008 (BTW)
- 10000062 (IVL)
- 310000000 (IVK)
- 310000003 (IVQ)
- 310000005 (IVE)

• Add the user_id parameter to SBO_SP_TransactionNotification

Starting from *SAP Business One version 10.0 FP 2108*, you can add user_id (integer type) as the 6th parameter into the procedure *SBO_SP_TransactionNotification*. Before you add the parameter, close all active applications connected to the SAP Business One database. This parameter is not applicable for *SBO_SP_PostTransactionNotice*.

Sample code for SAP Business One, version for SAP HANA

```

SQL
Alter PROCEDURE SBO_SP_TransactionNotification
(
    in object_type nvarchar(30),          -- SBO Object Type
    in transaction_type nchar(1),        -- [A]dd, [U]pdate, [D]elete, [C]ancel, C[L]ose
    in num_of_cols_in_key int,
    in list_of_key_cols_tab_del nvarchar(255),
    in list_of_cols_val_tab_del nvarchar(255),
    in user_id int -- Include only from version SAP Business One FP2108 and above
)
LANGUAGE SQLSCRIPT
AS
-- Return values
error int;          -- Result (0 for no error)
error_message nvarchar (200);  -- Error string to be displayed
begin

error := 0;
error_message := N'Ok';

-----
--  ADD YOUR    CODE    HERE

IF object_type = '17' and user_id = 1 THEN
BEGIN
    error := -1;
    error_message := 'You are not allowed to add/update salesorder';
END;
END IF;

-----

-- Select the return values
select :error, :error_message FROM dummy;

end;

```

Limitations:

- [3020664](#) - Transaction Notification Error Message Not Displayed When Project is Canceled.
- [1480546](#) - SBO_SP_PostTransactionNotice and SBO_SP_transactionnotification on Period and Volume Discounts
- [2559941](#) - Transaction Notification Stored Procedure Not Showing Custom Error Message
- [3019631](#) - Incorrect System Message for Blocked Transactions.
- [2856706](#) - Transaction Notification Error Message Not Displayed When Changing Production Order Status.
- [1452783](#) - Translations in UDO Child Table Disappear when Table Updated
- [2559941](#) - Transaction Notification Stored Procedure Not Showing Custom Error Message.
- [2219358](#) - SBO_TransactionNotification cannot be triggered when canceling an invoice
- [2047284](#) - Incorrect exceeded value of credit and commitment showed that on system message after transaction is prevented from adding by SBO_SP_TransactionNotification

Support Scope:

The *Transaction Notification* stored procedures are provided "as is" and the support does not apply for the logic implemented in the procedures.

The guidelines below should be followed when building the logic in the notification procedure.

1. Do not change ISOLATION LEVELS within the procedures.
2. Do not COMMIT/ROLLBACK transactions within the procedure.
3. Do not directly update/delete/insert SAP Business One tables within the procedures.
4. Do not use long, time-consuming logic because it may introduce performance issues.

Please refer to the [maintenance policy](#) document and KBA [3396055](#) for data modification guidelines.

This SAP Note is referenced in the document [What's New in Previous Versions of SAP Business One 10.0](#) and [SAP Business One 10.0](#).

Other terms

SDK, documentation, DI Event Service, approval, SBO_SP_TransactionSupport, SBO_SP_PostTransactionSupport, sql, @error, Internal Error 3621 occurred Message 13, CTNS

Products

Products
SAP B1 VERSION FOR SAP HANA all versions
SAP BUSINESS ONE all versions

该文档参考

SAP Note/知识库文章	组件	标题
3396055		SBO-SDK-DEV
3020664	SBO-SDK-TNSP	Transaction Notification Error Message Not Displayed When Project is Canceled.
3019631	SBO-SDK-TNSP	Incorrect System Message for Blocked Transactions
2856706	SBO-SDK-TNSP	Transaction Notification Error Message Not Displayed When Changing Production Order Status
2559941	SBO-SDK-TNSP	Transaction Notification Stored Procedure Not Showing Custom Error Message
2219358	SBO-SDK-TNSP	SBO TransactionNotification cannot be triggered when cancelling an invoice
2047284	SBO-SDK-TNSP	Incorrect exceeded value of credit and commitment showed on system message after transaction is prevented from adding by SBO SP TransactionNotification
1480546	SBO-SDK-TNSP	SBO SP PostTransactionNotice and SBO SP transactionnotification on Period and Volume Discounts
1452783	SBO-SDK-UDO	Translations in UDO Child Table Disappear when Table Updated
1298609	SBO-SDK-TNSP	UDO Error when updating user defined Table or Object

该文档被参考

SAP Note/知识库文章	组件	标题
3362634		<u>SBO SP PostTransactionNotice Does Not Work</u>